

Arnab Bhattacharjee

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Research Interests

Human-Computer Interaction (HCI), Agentic AI, Computer Science Education (CSEd), Human-AI Interaction, Systems

Education

Bangladesh University of Engineering and Technology (BUET)

Feb 2020 – Mar 2025

B.Sc. (Honours) in Computer Science and Engineering

CGPA: 3.88 / 4.00

Research Experience

Human-AI Co-Creation for Technical Lecture Content | HCI, Agentic AI, CS Education

Jan 2026 – Sep 2026

- Investigated how computer science educators author technical lecture content with generative AI tools.
- In a formative study, conducted semi-structured interviews with university educators and analyzed them using reflexive thematic analysis to identify their challenges and derive design goals.
- Guided by these goals, built an agentic pipeline that generates technical lecture content under the educator's direction.
- Conducted a qualitative user evaluation through focus group discussions (FGDs) with educators.

Language Effects on LLM Web Agents | Pilot research project, extension of undergraduate thesis

Jul 2026 – Sep 2026

- Evaluated an HTML-based LLM web agent across several widely spoken languages in two settings: a controlled offline benchmark in which the page structure stays fixed and only the language changes, and live task execution on real localized websites.
- Quantified how much of the agent's observation actually changes with the language, alongside task performance.
- Found that performance stayed close to English even when most of the page text was rewritten, while non-English pages consumed a substantially larger share of the model's context budget.

Evaluating the Multilingual Capabilities of LLM-based Web UI Agent | Undergraduate Thesis

Nov 2023 – Mar 2025

- Developed a multilingual web agent that navigates web pages and performs actions to complete complex, multi-step tasks given by a user.
- Fine-tuned a multilingual model (mT5-base) on an extended Mind2Web dataset, and built web agents on three frontier LLMs (Gemini-2.0-Flash, GPT-4o-mini, DeepSeek-V3) to evaluate their zero-shot performance.
- Built a VLM-based automated evaluation system for web agents.
- Supervised by [Dr. Rifat Shahriyar](#), Professor, BUET. [Extended abstract](#).

Poster

Abir Muhtasim*, **Arnab Bhattacharjee***, Abhik Bhattacharjee, Faria Huq, Rifat Shahriyar. *How well do LLM Web Agents work for non-English Language?* LLM Agent Workshop, Carnegie Mellon University, Pittsburgh, PA, USA. May 2, 2024. [Poster](#).

(*Equal contribution)

Academic and Professional Experience

United International University (UIU)

Jul 2025 – Present

Lecturer, Department of Computer Science and Engineering

Dhaka, Bangladesh

- Courses taught: Data Structures and Algorithms, Discrete Mathematics, Object-Oriented Programming, Operating Systems, Digital Logic Design, Network Security, and Social, Environment and Engineering Ethics.
- Course coordinator for Microprocessor and Microcontroller, including embedded systems projects with Arduino and Raspberry Pi.
- Trained in Outcome-Based Education (OBE), the framework of UIU CSE's Washington Accord accreditation, applied in designing course curricula and lesson plans.

Samsung Research and Development Institute, Bangladesh

May 2025 – Jul 2025

Software Engineer

Dhaka, Bangladesh

- Developed an Android application (Kotlin, Android Studio) for an intelligent pen that writes on any surface, with the strokes appearing on the Android device automatically.

Projects

- Machine Learning Assisted Gene Expression Restoration** | *Machine Learning, Bioinformatics* | [GitHub](#) 2024
- Explored nonlinear models for identifying the transcription factors that restore gene expression from a source cell state to a target state, extending the linear GAGER approach. Nonlinear models often reduced expression differences using fewer transcription factors across several scRNA-seq datasets.
- Project Tokkhok** | *React, FastAPI, PostgreSQL, Supabase, RAG* | [Demo](#) | [GitHub](#) 2025
- AI-assisted learning platform for Bengali-speaking students: chat with an uploaded knowledge base (retrieval over a vector database) in Bangla, Banglish (Bangla typed in English letters), or audio, and a smart note editor that converts Banglish notes to proper Bangla.
- Buldr-Bangladesh** | *React, Spring Boot, PostgreSQL, Tailwind CSS, Firebase Auth* | [GitHub](#) 2025
- Full-stack platform for engineers to learn (AI chatbot, smart notes, summarizing and paraphrasing tools), showcase projects, sell projects through a marketplace with order tracking, and get auto-generated portfolios.
- Deepfake Image Detector Robust Against Adversarial Attacks** | *PyTorch, ViT, YOLOv11* | [GitHub](#) 2025
- Detector for real and AI-generated images (CIFAKE). Feature dropping and augmentation reached up to 80% accuracy on adversarial data, and custom adversarial noise and dropout layers improved robustness.
- TechWorld** | *Node.js, Express.js, PostgreSQL, Oracle PL/SQL* | [GitHub](#) 2022
- E-commerce website for PCs with complex SQL and PL/SQL operations. Core feature: PC-BUILDER for assembling a custom PC from compatible parts.

Technical Skills

HCI Methods: Semi-structured interviews, reflexive thematic analysis, focus group discussions, formative studies

AI and LLMs: Agentic LLM pipelines, RAG, LLM-as-judge evaluation, PyTorch, fine-tuning transformer models

Languages: Python, C, C++, Java, JavaScript, Kotlin, SQL, Bash

Web and Databases: React, Tailwind CSS, Node.js, Express.js, Spring Boot, FastAPI, PostgreSQL, Supabase, Oracle DB

Tools: Playwright, Android Studio, Postman, Swagger, LaTeX

Honors and Awards

- Dean's List Scholarship, BUET (all four years)
- Champion, ITVerse 2023 "TechTales" Project Showcase Competition, University of Dhaka 2023
- Finalist, BUET CSE Fest Capture the Flag (CTF) Competition 2023
- Board General Scholarship, Secondary School Certificate (SSC) Examination 2017
- Talent Pool Scholarship, Junior School Certificate (JSC) Examination 2014
- Talent Pool Scholarship, Primary School Certificate (PSC) Examination 2011
- Competitive programming: 600+ problems solved, max rating 1519 on [Codeforces](#)

References

Available upon request.